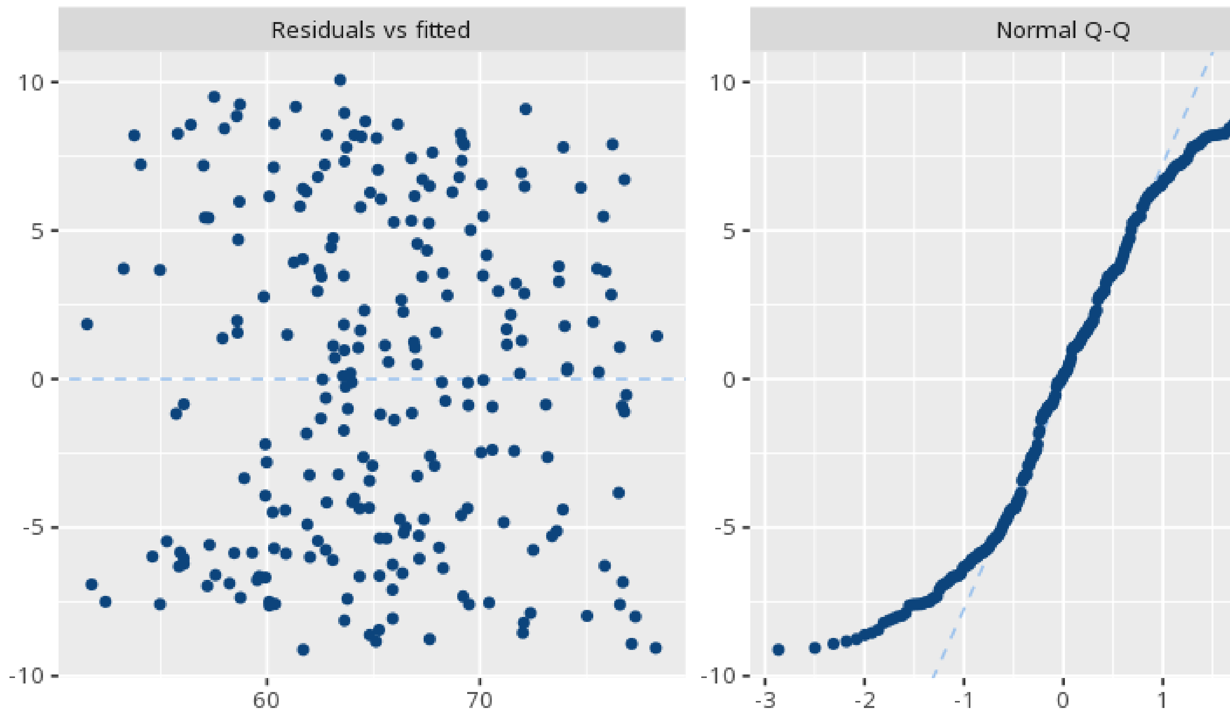
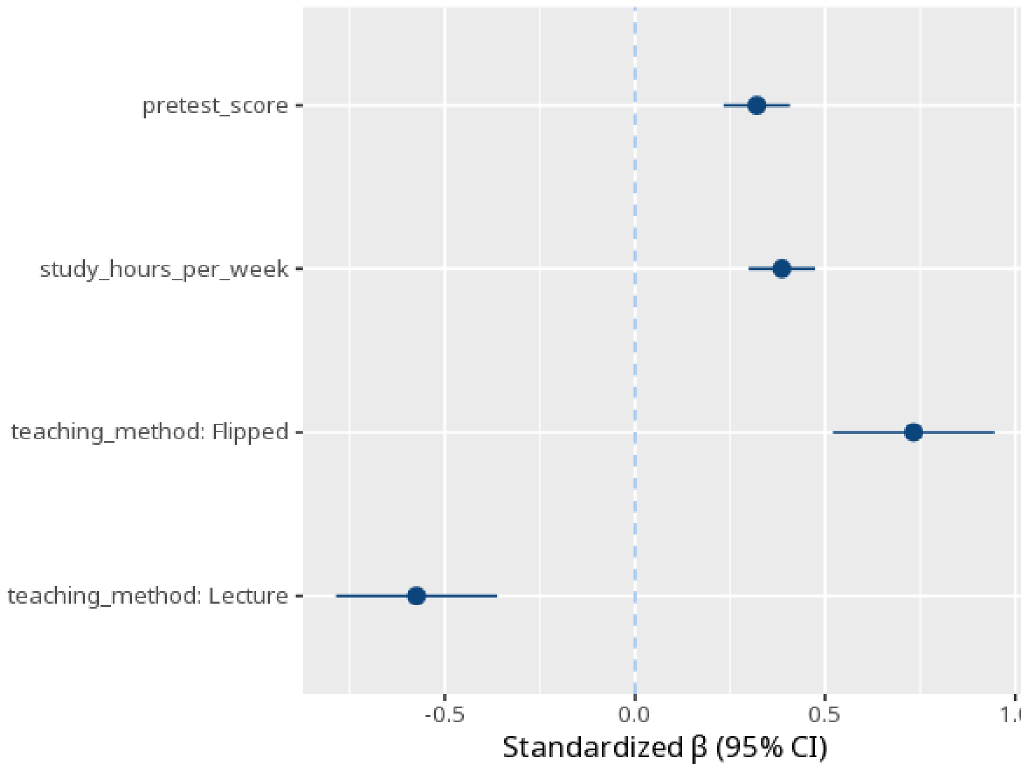
1 Multiple linear regression

	(1)	β	VIF
(Intercept)	41.21		
	(2.33)		
	[36.62, 45.81]		
pretest_score	0.31	—	1.01
	(0.04)		
	[0.22, 0.39]		
study_hours_per_week	1.30	—	1.02
	(0.15)		
	[1.01, 1.60]		
teaching_method: Flipped	5.97	—	1.00
	(0.88)		
	[4.24, 7.71]		
teaching_method: Lecture	−4.68	—	1.00
	(0.88)		
	[−6.41, −2.95]		
Num.Obs.	240		
R^2	0.55		
R^2 Adj.	0.54		
F	70.38		
RMSE	5.49		
AIC	1510.28		
BIC	1531.16		
Log.Lik.	−749.14		


Residuals vs fitted
Normal Q-Q
10
5
0
-5
-10
60
70
-3
-2
-1
0
1



Each predictor's B is its effect holding the others constant; p and CI assess it. R^2 is the joint variance explained; VIF flags predictors that overlap too much (multicollinearity). B is in each predictor's own units, so to compare which predictors matter most use the standardized coefficient (β), not B. Your significance threshold (α) is 0.05.

The model explained R^2=.55 of the variance, F(4,235)=70.38, $p < .001$; predictor pretest_score gave B=0.31, $p < .001$.

Statistical basis: Ordinary least squares regression coefficients (B), SE, CI (R Core Team (2026). R: A Language and Environment for Statistical Computing. R Foundation for Statistical Computing, Vienna.); Standardized coefficients (β) (Lüdecke D, Ben-Shachar MS, Patil I, Makowski D (2020). "Extracting, Computing and Exploring the Parameters of Statistical Models using R." Journal of Open Source Software, 5(53), 2445.); Variance inflation factor (VIF) cutoff guidance (OBrien, R. M. (2007). "A caution regarding rules of thumb for variance inflation factors." Quality & Quantity, 41(5), 673-690.).